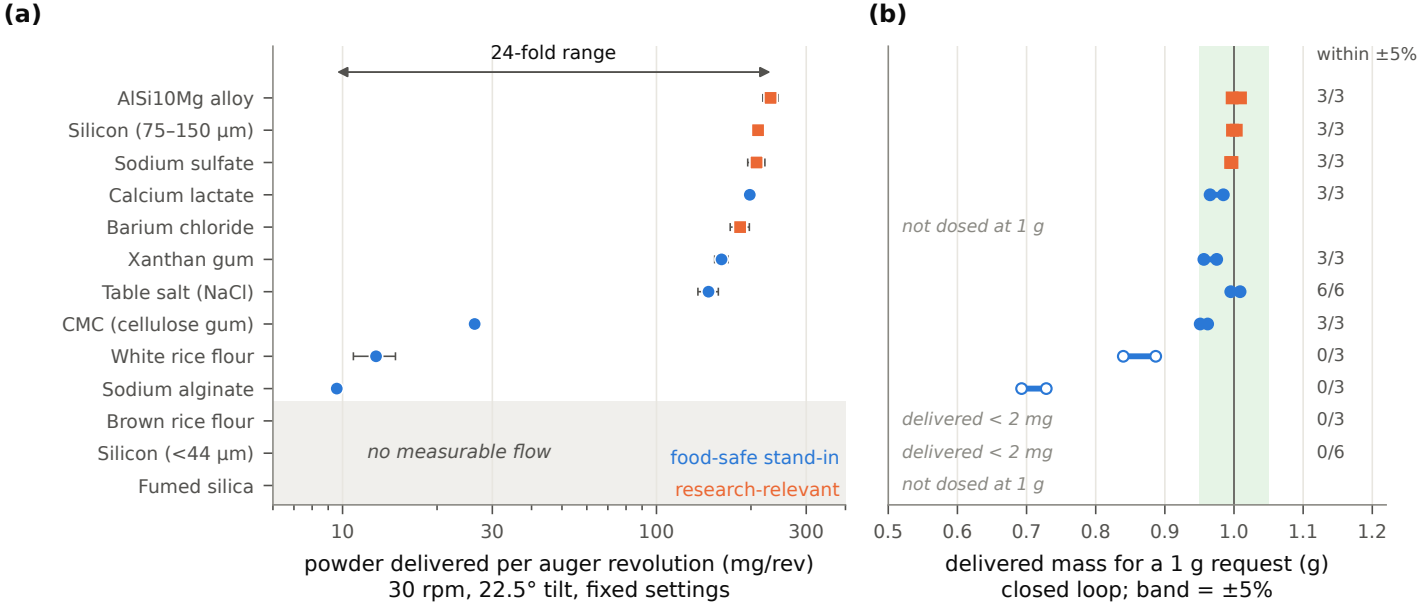
(a)
24-fold range
AlSi10Mg alloy
Silicon (75–150 µm)
Sodium sulfate
Calcium lactate
Barium chloride
Xanthan gum
Table salt (NaCl)
CMC (cellulose gum)
White rice flour
Sodium alginate
Brown rice flour
Silicon (<44 µm)
Fumed silica
no measurable flow
food-safe stand-in
research-relevant
10
30
100
300
powder delivered per auger revolution (mg/rev)
30 rpm, 22.5° tilt, fixed settings
(b)
within ±5%
3/3
3/3
3/3
3/3
not dosed at 1 g
3/3
6/6
3/3
0/3
0/3
delivered < 2 mg
0/3
delivered < 2 mg
0/6
not dosed at 1 g
0.5
0.6
0.7
0.8
0.9
1.0
1.1
1.2
delivered mass for a 1 g request (g)
closed loop; band = ±5%